

Resource Plan - Splitly Project

1. Introduction

This document defines and allocates the necessary resources (human, tools, and equipment) to execute the **Splitly** project - a smart expense sharing and management application. The project is undertaken by a team of 6 third-year students from the University of Science (HCMUS), with the goal of building an optimized product starting from a manual entry workflow and evolving into an AI-integrated (receipt scanning) solution.

2. Human Resources: Roles & Responsibilities

The development team consists of 6 members. Given the nature of a student project, roles are distributed flexibly with cross-functional support. The specific roles are as follows:

Role	Quantity	Main Responsibilities
Project Manager (PM) / BA	1	Manages overall progress, distributes tasks, and coordinates resources. Primarily responsible for business documentation, User Stories, and workflows.
UI/UX Designer / Frontend Dev	1	Designs wireframes, mockups, and prototypes in Figma. Builds the frontend interface based on the design, ensuring a smooth UX for the user journey.
Frontend Developer	1	Focuses on user interface (UI) development, integrating split bill logic on the client-side, and connecting with Backend APIs.
Backend Developer / AI Engineer	1	Designs the Database and develops core APIs. Responsible for researching and integrating the OCR model (PaddleOCR/Tesseract) for receipt scanning.
Backend Developer	1	Supports API development, handles complex algorithms such as debt clearing logic, VietQR integration, and security measures.

Role	Quantity	Main Responsibilities
QA / Tester / DevOps	1	Creates test cases and executes testing (Manual/Automation). Assists in setting up the CI/CD environment and deploying the application.

3. Resource Allocation & Effort Estimation

Working Hours:

Each member commits to dedicating approximately **4 days/week** to the project (estimated at 16 - 24 hours/week/person, depending on the university schedule).

Weekly Effort Estimation:

- **Total person-days per week:** 6 members × 4 days/week = **24 person-days/week**.
- The following is the estimated Effort Distribution and resource priority across the main phases of the project:

Phase	BA / Design	Frontend	Backend & AI	QA & DevOps	Work Focus
1. Initiation & Analysis	High	Low	Low	Low	Finalize Requirements, DB & Figma design.
2. Core Dev (Manual Entry)	Low	High	High	Medium	Code core splitting logic, create APIs, assemble UI.
3. Advanced Dev (AI OCR)	Medium	High	High	Medium	Integrate receipt recognition (OCR) and VietQR API.
4. Testing & Refinement	Low	Medium	Medium	High	Fix bugs, UAT, evaluate AI accuracy.

4. Development Environment & Tools

The project fully utilizes modern tools to increase team productivity, particularly with AI assistance:

- **AI Coding Assistants:** The team is equipped with **3 Codex accounts** to assist with coding. These accounts will be rotated or prioritized for 3 Developers (Frontend and Backend) during intensive coding phases to accelerate writing logic, generating test cases, and handling complex bill-splitting algorithms.
- **Project & Code Management:** GitHub (source code management, Issues, PRs), Notion/Trello (task management), Discord/Zalo (Communication).
- **Design:** Figma (Prototype and UI assets).
- **Core Tech Stack & AI:** React/Vue (Frontend), Node.js/Python (Backend & AI Services), PaddleOCR/Tesseract (OCR receipt recognition).

5. Resource Schedule

- **Initial Phase:** Resources are heavily focused on the Project Manager, UI/UX Design, and setting up the Backend/Database architecture. (The 3 Codex accounts can be used to generate boilerplate code).
- **Middle Phase (Core Development):** Frontend and Backend resources are operating at 100% capacity. The 3 Codex accounts are used continuously to accelerate the completion of the *Manual Entry* flow and *Debt Clearing* logic.
- **Final Phase (AI Integration & Testing):** Backend/AI focuses on OCR image processing and data extraction. The Tester operates at maximum capacity to evaluate AI accuracy and build fallback logic for cases where OCR misreads data.

6. Resource Risks & Mitigation

Risk	Impact	Mitigation Strategy
Schedule conflicts with exams/other assignments reducing actual work days (below 4 days/week).	High	Plan with buffer time. Accelerate progress during free weeks. Maximize the use of Codex to compensate for slower coding speeds during busy times.
Conflicts when sharing 3 Codex accounts among 4-5 developers.	Low	Establish a clear shift schedule (morning/afternoon/evening) or use screen sharing (Pair-programming) so 2 devs can share 1 account.
Coding Bottlenecks.	Medium	Organize Pair-programming (1 person dictates the algorithm, 1 person prompts Codex to generate code quickly).

Risk	Impact	Mitigation Strategy
Lack of in-depth knowledge in AI/OCR integration leading to excessive research time.	High	Allocate the AI Engineer to build Proof of Concepts (PoCs) from the very first week. Use Codex to ask for architectural suggestions for OCR.